Comparative Study of Embedded Boards

What You Must Do

You must:

- Understand the technical specifications
- Identify similarities and differences
- Analyze strengths and weaknesses
- Justify which board is better depending on the application

Board Assignment

Each student has a different pair:

Pair	Board A	Board B	Student
1	Arduino Uno	Arduino Mega 2560	
2	Arduino Nano	Arduino Uno	
3	Arduino Mega 2560	ATmega328P	
4	ESP8266	ESP32	
5	ESP32	Arduino Uno	
6	STM32 Nucleo	Arduino Uno	
7	Raspberry Pi Pico	Arduino Nano	
8	Raspberry Pi Pico W	ESP8266	
9	STM32 Nucleo	ESP32	
10	Raspberry Pi 4	Orange Pi	
11	Raspberry Pi 4	Banana Pi	

Presentation Format

- Individual work
- 5–6 slides maximum
- 6–8 minutes oral presentation
- Simple, clear and technical English

Required Slide Structure

Slide 1 – Introduction

- Presentation of both boards

- Category (MCU / SBC)
- Main specifications
- Typical applications

Slide 2 – Hardware Comparison (Table Recommended)

Compare:

- Processor (type + frequency)
- RAM / Flash
- Number of GPIO
- Communication interfaces
- Power supply
- Power consumption

Slide 3 – Software & Development

Compare:

- Programming languages
- IDE
- Ease of use
- Community support

Slide 4 – Use Case Analysis

Choose one application.

Explain:

- Which board is more suitable
- Why (technical justification required)

Slide 5 – Cost & Practical Aspects

- Price
- Availability
- Accessories needed

Slide 6 – Final Conclusion

- Strengths of Board A
- Strengths of Board B
- Final recommendation depending on the project